

SON MANH TRAN

Hanoi, Vietnam

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WORK EXPERIENCE

Vietcombank – Quant Department, Market Risk Team

Jun 2025 – Present

Quantitative Analyst

Hanoi, Vietnam

- Own end-to-end pricing and risk engines across 6 asset classes – interest-rate derivatives, FI bonds, US Treasuries, commodities, FX options, and loan IROs – calibrated daily to live market data and used across the bank's market risk book
- Built counterparty credit add-on exposure models for FX and derivative positions, underpinning regulatory capital calculations under the bank's Basel-aligned market risk framework
- Led replacement of legacy Power Automate workflow with a centralized database, saving the team 20 analyst hours per week and materially improving data processing speed
- Automated trade-position and market-data feeds from FIS Front Arena into downstream pricing and risk systems, eliminating manual uploads and reducing operational risk

University of Texas at Austin

Jun 2024 – Apr 2025

Graduate Research Assistant

Austin, TX

- Designed machine learning algorithms for high-dimensional PDEs applied to engineering and uncertainty quantification, advised by Prof. Tan Bui-Thanh
- Published research (USNCCM18, 2025) demonstrating that transformer architectures implicitly learn classical numerical solvers for PDEs – bridging deep learning and applied mathematics
- Developed score-based generative models grounded in a novel convexified-PDE formulation; second paper accepted at USNCCM18 (2025)

University of Texas at Austin

Jun 2024 – Apr 2025

Teaching Assistant

Austin, TX

- Led class sessions and office hours for undergraduate engineering students; responsible for delivering course material and assessing student work
- Supervised groups of undergraduates conducting research projects under Prof. Bui-Thanh's lab, providing technical direction and feedback throughout the research cycle

Vietcombank

Aug 2022 – Dec 2022

Quantitative Risk Intern

Hanoi, Vietnam

- Redesigned the corporate credit Early Warning System (EWS), improving identification of at-risk borrowers ahead of default events
- Built Basel II-compliant statistical risk models for low-default portfolios, supporting the bank's regulatory reporting cycle
- Delivered data analysis and modeling in SAS, MATLAB, SQL, and Python

EDUCATION

University of Texas at Austin, Ph.D. program in Engineering Mechanics

Jun 2024 – Apr 2025

Advisor: Prof. Tan Bui-Thanh

Research: Machine Learning, Numerical Analysis, Uncertainty Quantification

Program discontinued following 2025 U.S. federal research funding cuts.

Knox College, B.S. in Mathematics

Sept 2019 – Jun 2024

Thesis: *The Estimation Error of PCA in Infinite Dimension*

Lam Son High School for the Gifted, Specialized in Mathematics

Sept 2015 – Jun 2018

TECHNICAL SKILLS

Quantitative	Derivative pricing, numerical methods, Monte Carlo simulation, model calibration, market risk, counterparty credit risk
Programming	Python (JAX, PyTorch, NumPy/SciPy), SQL, MATLAB, C/C++, R, SAS, L ^A T _E X
Domain	Basel II, FIS Front Arena, financial markets, regulatory capital frameworks
Languages	Vietnamese (native), English (professional)

SELECTED PROJECTS

Neural Networks for Rational Option Pricing	Jun 2022 – Aug 2022
· Embedded no-arbitrage constraints directly into neural network architecture for derivative pricing, ensuring economically consistent model outputs · Benchmarked ensemble neural network pricing against market option-price datasets · Link to presentation	

PUBLICATIONS AND CONFERENCE PRESENTATIONS

Convexifying PDEs as basis for designing score-based generative models	2025
Alois Duston, Son Tran , Nirmal Patel, Tan Bui-Thanh <i>U.S. National Congress on Computational Mechanics (USNCCM18)</i>	
Transformers are secretly numerical solvers	2025
Hai Nguyen, Son Tran , Venugopal Ranganathan, Tan Bui-Thanh <i>U.S. National Congress on Computational Mechanics (USNCCM18)</i>	
Herd behavior in Vietnam’s stock market: Impacts of COVID-19	2023
Hong Mai Phan, Thi Nhu Quynh Le, Vu Duc Hieu Dam, Manh Son Tran , Thi Hoai Linh Truong, Quoc Anh Le <i>Cogent Economics & Finance</i>	

HONORS AND AWARDS

UT Austin Aerospace Engineering Department Fellowship	2025
Knox College Mathematics Department Awards (top student)	2020 – 2024
Knox College Full-Tuition Scholarship (four years)	2019 – 2024